## Experiment 2 & 3: Introduction to Exploratory Data Analysis and Descriptive statistics using R

******REPLACE THE "df" WITH REQUIRED DATAFRAME**

**For Creating a small Dataset**

```
data <- data.frame(
  age = c(22, 25, 29, 30, 35, 40, 45, 50),
  income = c(30000, 35000, 45000, 50000, 60000, 70000, 80000, 90000),
  education = c("B", "B", "M", "M", "M", "D", "D", "D")
)
```

### Load and inspect data

```
data <- read.csv("sample_data.csv")
head(data)
str(data)

sapply(df, class) # data type of each column

sum(duplicated(df)) # duplicate rows
```

1. ### Basic summaries

```
summary(df)

mean(data$age) .
median(data$income)
sd(data$age)
```

2. ### Simple plots

```
# Histogram
hist(data$age, main = "Age Distribution", xlab = "Age")

# Boxplot
boxplot(data$income, main = "Income Distribution")

# Scatter plot
plot(data$age, data$income,
    xlab = "Age", ylab = "Income",
    main = "Age vs Income")
```

3. **Group-wise summaries**

```
library(dplyr)
data %>%
 group_by(education) %>%
 summarise(
  mean_income = mean(income),
  median_age = median(age)
 )
```

## Descriptive Statistics (Mean, Median, Variance, Correlation)

## Objective

- Compute and interpret descriptive statistics.
- Understand correlation between variables.

## Sample Data

Use same dataset as before

```
data <- read.csv("sample_data.csv")
```

## Procedure

1. **Mean and Median**

```
mean_age <- mean(data$age)
median_income <- median(data$income)
```

2. **Variance and Standard Deviation**

```
var_age <- var(data$age)
sd_income <- sd(data$income)
```

3. **Correlation**

```
cor_age_income <- cor(data$age, data$income)
cor_matrix <- cor(df)

round(cor_matrix, 2)
```

```
library(corrplot)
corrplot(cor_matrix, method = "color", type = "upper",
         tl.col = "black", addCoef.col = "black")
```

4. **Covariance (optional)**

```
cov_age_income <- cov(data$age, data$income)
```

5. **Print results**

```
cat("Mean age:", mean_age, "\n")
cat("Median income:", median_income, "\n")
cat("Variance of age:", var_age, "\n")
cat("Std dev of income:", sd_income, "\n")
cat("Correlation(age, income):", cor_age_income, "\n")
```

6. **Interpretation**

   - Mean: average value.
   - Median: middle value (robust to outliers).
   - Variance: spread of data.
   - Correlation: strength and direction of linear relationship.
   - If `cor_age_income` is close to +1, strong positive linear relationship.

7. **For Visualization**

```
plot(x, y, main = "Scatter Plot of x vs y",
     xlab = "x", ylab = "y", pch = 19, col = "blue")
abline(lm(y ~ x), col = "red")  # for adding regression line
```